

Lorenzo Veronese

Curriculum Vitae

Como, Italy

✉ lorenzo.veronese@polimi.it

🌐 <https://lorenzoveronese.github.io/>

in www.linkedin.com/in/lorenzo-v-veronese/

ORCID: 0000-0001-9204-3363

Profile

PhD Candidate at Politecnico di Milano, specializing in Computational Cognitive Neuroscience and Neuro-inspired AI. My research focuses on developing state-of-the-art AI models, grounded in fMRI data, to emulate complex human functions such as planning (through theory-driven AI models informed by neuroscience, psychology, and neuropsychology) and visual perception (via advanced two-stage generative decoding architectures). Published work includes enhancements to neural decoding pipelines, demonstrating superior performance and efficiency (more than 95% improvement) for real-time applications.

Education

- 2024–Present **Ph.D., Bioengineering**, *Politecnico di Milano (Milan, Italy)*, Supervisor: Prof. Pietro Cerveri, Computational Cognitive Neuroscience for Understanding Human Planning and Perception.
- 2022–2024 **M.Sc., Computer Science Engineering, Artificial Intelligence**, *Politecnico di Milano (Milan, Italy)*, Thesis: "Visual stimulus reconstruction from BOLD fMRI signal using generative AI", Final grade: 110 *cum laude*/110 (GPA: 29.2).
- 2019–2022 **B.Sc., Computer Science Engineering**, *Politecnico di Milano (Milan, Italy)*.

Research Experience

- 2024–Present **Research fellow and PhD student**, *Politecnico di Milano (Department of Electronics, Information and Bioengineering) and IRCCS Ospedale San Raffaele (Department of Neuroradiology)*.
- Neuro-AI Generative Modeling: Architecting a novel two-stage generative decoding pipeline to reconstruct complex visual perception from fMRI data.
 - Computational Cognitive Neuroscience: Formulating a theory-driven probabilistic model of human planning.
 - Machine Learning in Neuroscience: Design of machine learning analysis pipelines for clinical neuroscience datasets.
- 2021–2023 **Student Researcher**, *Politecnico di Milano (Department of Electronics, Information and Bioengineering), NECSTLab*.
- Generative AI for Data Harmonization: Engineered a CycleGAN-based deep learning framework for the harmonization of histopathological images across different imaging domains.
- 2022–2023 **Student Researcher**, *Politecnico di Milano (Department of Electronics, Information and Bioengineering), DEEP-SE Lab*.
- Formal Modeling: Modeled the dynamics of psychotherapy conversations using Stochastic Hybrid Automata and natural language processing.

2021–2022 **Student Researcher**, *Politecnico di Milano (Department of Electronics, Information and Bioengineering)*, AIRLab.

- Robotic Design and Prototyping: Led the full-stack design (mechanical and software) of an assistive robotic device to support individuals with motor disabilities in writing tasks.

Peer-reviewed Scientific Publications

Veronese, L., Pecco, N., Moglia, A., Scifo, P., Castellano, A., Mainardi, L., Della Rosa, P.A., & Cerveri, P. (2025). Evaluating Semantic Brain Regions Contribution to Visual Neural Decoding Performance on Different Classes of Stimuli *20th conference on Computational Intelligence methods for Bioinformatics and Biostatistics (CIBB 2025)*.

Veronese, L., Moglia, A., Pecco, N., Della Rosa, P.A., Scifo, P., Mainardi, L., & Cerveri, P. (2025). Optimized AI-based neural decoding from BOLD fMRI signal for analyzing visual and semantic ROIs in the human visual system. *Journal of Neural Engineering*, 22(4).

Veronese, L., Poles, I., D'Arnese, E., Santambrogio, M.D. (2023). Stain Transfer using CycleGAN for Histopathological Images. *IEEE EUROCON 2023-20th International Conference on Smart Technologies*.

Teaching

2025 **Teaching Assistant, Neuroengineering**, *Politecnico di Milano*

- Delivered lectures on deep learning architectures and methodologies.
- Supervised student groups in designing and training end-to-end deep learning models for decoding visual categories from fMRI data.
- Co-designed course materials and organization.

2025 **Teaching Assistant, Fundamentals of Computer Science**, *Politecnico di Milano*

- Led laboratory sessions for undergraduate students, teaching algorithmic thinking and data structures in C programming.
- Co-designed course materials.

Awards & Honors

2025 Selected by Nova talent (ID: 6698IITXR XG57SY19CHV5KD3E) — Nova

2022 Scholarship for academic merit — Town of Cucciago & Rubelli

Skills

Expertise	Computational Cognitive Neuroscience, Artificial Intelligence, Machine Learning, Deep Learning, fMRI, Cognitive Neuroscience, Public Speaking
Programming	Python, PyTorch, TensorFlow, MATLAB, Bash
Tools	Git, LaTeX, Statistical Parametric Matching (SPM), FSL
Languages	English (fluent), Italian (native)

References

Prof. Pietro Cerveri Full Professor — Università di Pavia
pietro.cerveri@unipv.it

Dr. Pasquale Anthony Della Rosa Scientific Collaborator — IRCCS Ospedale San Raffaele
dellarosa.pasquale@hsr.it